# Data Modelling Assignment

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***Abstract*—The report introduces the QSAR dataset and its importance. It provides information ad the steps taken during the building of a predictive model for the dataset for biodegradability and confirming it using cross validation. The steps included data visualisation like heat maps and violin plots, to data processing techniques like PCA and machine learning algorithm like Random Forest Classification and Random Forest Regression and comparing which is a better modelling algorithm for the given problem The lease mean squared error was observed to be close to 0 and a confusion matrix was also developed to understand how many predictions are correct and incorrect per class.**

## I. Introduction

The Quantitative Structure-Activity Relationship (QSAR) model was developed for assessing biodegradability of chemicals and at the same time it acted as an alternative for animal testing [1]. This report aims to build a predictive model for the QSAR dataset to get the prediction for biodegradability using Random Forest Classifier which is different from the ones that were used in [1] where they used models like k-Nearest Neighbours(kNN), partial least squares discriminant analysis, Support Vector Machines(SVM) along with their consensus models. There have been many more models like Graph Convolution Network(GCN) [2] which had been a more straightforward and stable way of modelling the data set, or a combination of kNN, SVM and Regulated Logistics Regression (RLR) [3], here the method used is Random Forest Classifier. There are several advantages to using this model over any other models. Some of them are:

1) High Accuracy
2) Robust to Over fitting
3) They work well for high dimensional data
4) They have reduced variance

Data modelling in general come with many ethical challenges with one of the major ones being biasness. As explained in the article [4], the bias that creeps in the model can lead to various issues during recruitment, wrong facial recognition in many places which can exhibit racism as well. Even for the QSAR data set bias may occur and there has to be regular updates and monitoring which are needed to be done.

## II. Data Processing

The QSAR dataset provided was first visualised using a heat-map and box plot. The resultant of both are shown in Fig 1 and Fig 2 respectively. Heat maps are a really good way of identifying areas of similarities between two columns. In the case of the QSAR data set, the heat map gives a relation between each feature for one chemical. The colors shown on the heat map indicate that each feature of the chemical is related to one another and hence is not a good way to visualise

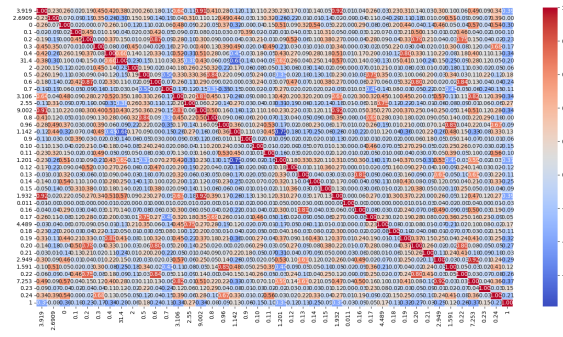

Fig. 1. Heat Map for QSAR Dataset

the data which brings us to the violin plot in Fig 2. The Box Plot shows how the values are dispersed and the skewness of the dataset while also giving information about the data in a particular column. It also shows the outliers within a dataset which provides an understanding of the next steps needed to be taken for further data processing which involves the removal of outliers.

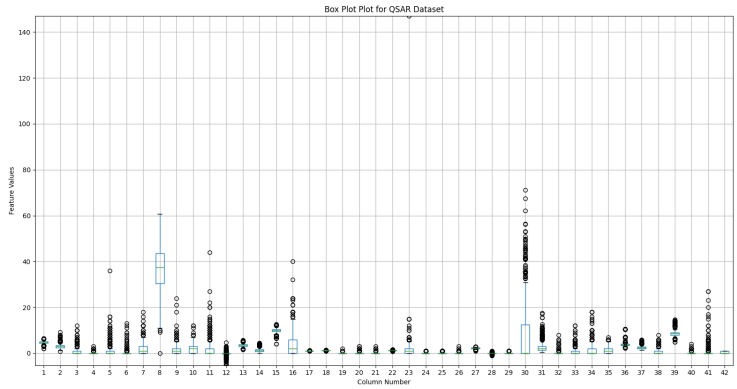


Fig. 2. Box Plot for QSAR Dataset

For the removal of outliers, let's first take the data set and remove the duplicate rows which makes sure there are no biased results as it can distort statistical measures. Once the duplicated data is removed, standardization of the new dataset takes place to normalize the sacles in the data and to avoid numerical instability. Now comes the step for removing the outliers as discussed. Outliers had been removed because they can make the model robust, give reduction in terms of predictive accuracy, can play a part in overfitting of data and provide misleading conclusions. The outliers are removed based on z-score values. In this method the mean value and

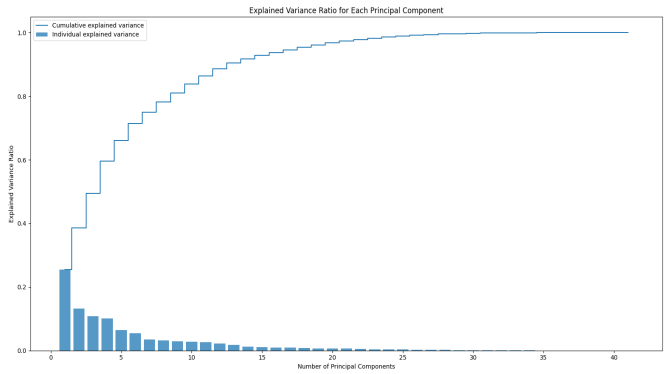


Fig. 3. Explained Variance Plot for QSAR Dataset

standard deviation of each column are calculated. The formula to calculate the z-score is given by:

$$zscore = (dataPoint - meanValue)/standardDeviation$$

Once the z-score for every data point in a column is calculated, the entire dataset is filtered and a new dataset is created which doesn't include rows with z-scores above a threshold which is generally considered to be 3.

The next step involves the process of dimensionality reduction through a process known as PCA. This reduction technique helps in redcuing the dimensions of a detasest while still maintaining the originality in the variables to a great extent. For a high-deimensional data like QSAR dataset, reducing the dimensions allows more efficiency and less computationally expensive models. It also filters out the unwanted noise and also helps in removal of redundant dataset. The process of PCA is done in 2 steps; the first step involves the making of a explained variance ratio plot. This provides with a value for the number of components which will be used for the second step. The plot for the explained variance ratio of the dataset obtained after removing outliers is as shown in Fig 3. Once the plot is created, the number of components chosen depends upon the elbow point after which adding more components will not increase the explained variance. In this case, a cumulative explained variance was selected with a threshold for 99% making sure that almost most of the variance is maintained. The second step is using the number of components obtained from the plot and applying it into the PCA algorithm to get a new dataset with reduced dimensions.

For the obtained dataset after removal of the outliers, during step 1, the number of components obtained was 26 which was then used in step 2. On the completion of step 2, a new data set is created with a new dimension of 26 datapoint and one column which will be the target variable, biodegradability.

## III. Methodology

The new dataset after going through the data processing methods as explained above, has to go through a final step which includes building of the machine learning algorithm. For this report, two models have been chosen which are the Random Forest Classification and Random Forest Regression.

TABLE I
Difference between Random Forest Classifier and random Forest Regression

| Random Forest Classifier | Random Forest Regression |
|---|---|
| 1. Useful for conditions when target variables are discrete | 1. Useful for conditions when target variables are continuous |
| 2. The output provides a class to which the datapoint belongs | 1. The output provides a numerical value |

Although both the models are an example of supervised learning they have a few differences in terms of their functioning which are indicated by the Table I.

They also have certain similarities like the fact that both the modelling algorithms build multiple decision trees during the training process and both of them reduce overfitting by using random feature selection.

### A. Random Forest Classifier

The steps involved in a random forest classification process are majorly distributed into two setps; one is the training and the other one is prediction

1) *Training*: training is done by building a collection decision trees whose nodes are nothing but random subsets of features which will be used for splitting
2) *Prediction*: Prediction is obtained from each tree and an aggregate is taken by using the majority vote and this allows to detect the final predicted class

### B. Random Forest Classifier

The steps involved in a random forest classification process are majorly distributed into two setps; one is the training and the other one is prediction

1) *Training*: training is done by building a collection decision trees whose nodes are nothing but random subsets of features which will be used for splitting
2) *Prediction*: Prediction is obtained from each tree and an aggregate is taken by using any averaging (mean or median) method and this allows to detect the final predicted class

The descriptions about the process shows that the main focus of a regression model for the random forest is focusing on prediction of continuous target variables which can be seen in the way that the aggregate is taken.

A k-fold cross validation was also done for both the modelling assists in mitigating the risk of overfitting. This approach contributes to a more dependable evaluation of the model's anticipated performance on data it has not encountered before. We have done a 5-fold cross validation to test each of the model separately. A confusion matrix was also created of each of the to understand how accurate were the models.

## IV. Model Analysis

### A. Random Forest Classifier

The random Forest classifier training progress plot as shown in Fig 6 gives us the information that the more number

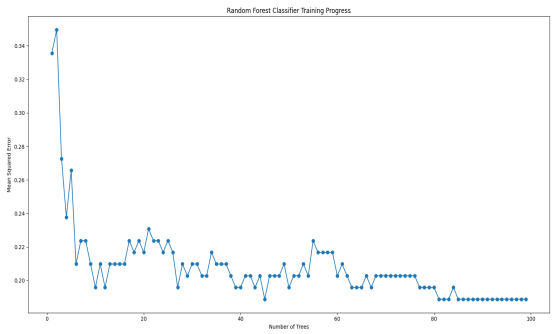


Fig. 4. Training Progress for Random Forest Classifier

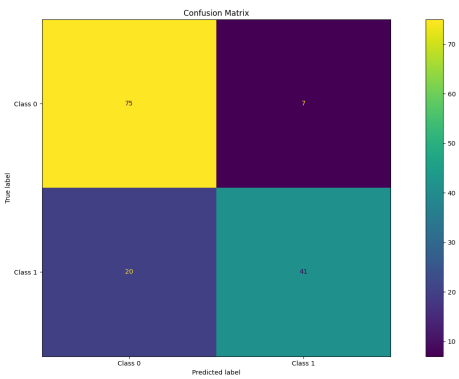


Fig. 5. Confusion Matrix for Random Forest Classifier

of trees are generated the mean squared error reduces. At the $99^{th}$ tree it is seen that mean squared error 0.189.

The Confusion Matrix created as shown in Fig 5 allows us to calculate the accuracy for the model. The true positive given in column 1 at the bottom gives a value of 41 while the false positive value was given as 7. the true negative value in column 0 on the top was given as 75 whereas the false negative value was given 20. This shows that 75 data points were read correctly as 0 (means not biodegradible) and 7 of them were incorrectly classified as positive. For the 41 truly positive values there were 20 values which were falsely classified as negative.

The accuracy from the set of values can be calculated by the formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP = True Positive; TN = True Negative; FP = False Positive; FN = False Negative

The Accuracy for the model created by us after replacing the values obtained in the above equations gives us a value of 0.811 which tells us that the model was 81.1% correct. Upon the 5-fold cross validation we get a set of 5 values which are = $[0.87412587, 0.85915493, 0.795774650.76760563, 0.84507042]$ letting us know how well the model performs and summarizes the skill of the model.

### B. Random Forest Regression

The random Forest classifier training progress plot as shown in Fig 6 gives us the information that the more number of trees are generated the mean squared error reduces. At the $99^{th}$ tree it is seen that mean squared error 0.119.

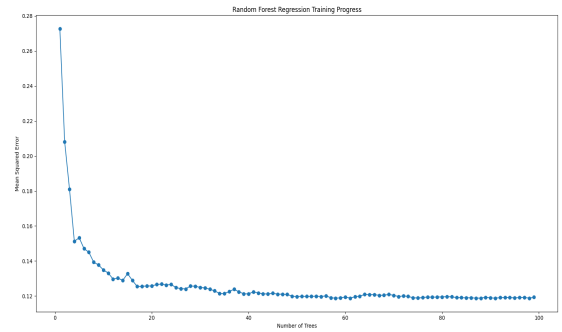


Fig. 6. Training Progress for Random Forest Classifier

Since the confusion matrix cannot be created for a Random Forrest Regression The only way to see the trueness of a model are through use metrics like R-squared ($R^2$) and mean-squared error (MSE). Since the mean squared error is lareayd given by ht eplot, let's see the r-squared value. The value of 0.512 gives us an understanding that model is able to explain the variability shown by the data around the mean. Since it is approximately halfway through 1, indicates the fact that atleast half of the dependant variables are predicted from independant variables and that the model is a decent fit for the data. If it was closer ot 0, it would have meant that the relationship between dependant and independant variables is non-linear.

The 5-fold cross validation for the Regression model gave the values: $[0, 0.4182287, 0, -0.65643262, 0.49550153]$. the negative values and 0 value indicate that the model is either overfitting the data or that the model is performing poorly.

## V. Conclusion An Recommendation

By seeing the analysis of both the Random Forest Classification and Random Forest Regression, we see that the mean squared error produced by both the models are pretty low, almost close to 0, whihc means that both the models are a decent fit for the data. We can't calculate the accuracy for a regression model, hence the only way to compare both of them wihtout any understanding is by seeing the mean-squared-value. The mean squared value for the regression model is 0.119 whereas the one for classification model is 0.189. This results in the regression model being a better fit for the QSAR dataset. But since the R-squared value and the results from a 5-fold cross validation is not promising, it can be concluded that Random Forest Regression is not the ideal algorithm to predict the biodegradability of a chemical. For the mean-squared error that occurs during the Random Forest Classification, it is seen that although it is a little higher than the one Regression produced, it is still relatively low and can be used for predicting the biodegradability for a chemical. This

understanding is also confirmed by the accuracy of the model which is 81.1% and all the values of 5-fold cross validation which are all positive and thus indicating that this model is a better fir for the data.

In general we can say that for a problem where the target value is distributed as similar as the one indicated by the biodegradability (i.e either 0 or 1; or any other distinct values), it is better to use a Classification technique rather than using a Regression technique as Classification best fits the data and has a higher chance of giving a better accuracy.

## VI. References

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[3] Alaa M. Elsayad, Medien Zeghid, Hassan Yousif Ahmed and Khaled A. Elsayad, "Exploration of Biodegradable Substances Using Machine Learning Techniques", Sustainability 2023, 15(17), 12764

[4] Niral Sutaria (CISA) (ACA), "Bias and Ethical Concerns in Machine Learning" Date Published: 29 August 2022 (Article)